

## Contact

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(Other)

## Top Skills

Session Initiation Protocol (SIP)  
SIP Trunking  
Big Data

## Languages

Tamil (Elementary)  
Hindi (Native or Bilingual)  
German (Elementary)  
English (Native or Bilingual)  
French (Elementary)  
Kannada (Limited Working)

## Certifications

AWS Cloud Solutions Architect  
ISC2 Systems Security Certified  
Practitioner (SSCP)

## Honors-Awards

Best Engineering Design & Fastest  
Acceleration  
Finalist (Top 30 out of 13K+)  
Best Paper Award  
Fr. Peter Mermier Scholarship

## Publications

Technical Validation of a Multimodal  
Emotion - Adaptive Biofeedback  
System for Autonomic Regulation  
Physics-Informed Stochastic  
Receding Horizon Control for  
Autonomous Energy Management in  
Solar Racing

# Pulkit Kumar

Voice AI & Platform @ Fettle | HMI & Big Data @ SolarMobil | 3+  
Patents & Springer Nature Published Author  
Bengaluru, Karnataka, India

## Summary

Shipped my first website and PCB at the age of 9. Today, I am a pre-final year undergrad building on the intersection of high-performance multi-agent systems and deterministic edge hardware. My work is divided into two core domains:- AI-native Distributed Systems: I architect agentic workflows and production-grade inference layers that serve 250K+ users. From implementing cross-modal recommender engines for wearables to deploying fault-tolerant ticketing systems handling 10x burst loads, the focus is on high-throughput reliability.- Safety-Critical Embedded Systems: As the Electronics Head for the university's solar racing team, I engineered the firmware for an Arm Cortex M7-based custom ECU. I've filed 3+ patents for adaptive EV interfaces and long-range telemetry systems, achieving a 55% reduction in thread latency for real-time control. Stack Highlights: AI/Software: LangGraph, PyTorch, Redis, Kubernetes, LiveKit, and Distributed Microservices. Embedded/ Hardware: MISRA C/C++, FreeRTOS, CAN Bus, FPGA (Verilog), and Edge AI (TensorRT). I am driven by an obsession for performance- whether I'm optimizing a digital twin for an endurance race or scaling cloud-native services to 99.99% uptime. If you are building high-stakes distributed/embedded systems, or anything on the edge- I'd love to connect!

## Experience

Fettle  
SDE (AI)  
January 2026 - Present (6 months)  
Hyderabad

Building voice AI for hospital operations & CX, and recommendation systems for corporate healthcare

Guaq AI  
Co-Founder

## Patents

A System and Method for Adaptive Interface in Electric Vehicle Control and Monitoring Applications

A System and Method for Long-Range Telemetry and Observability for Solar Electric Vehicles

A System and Method for Automated Robotic Solar Panel Cleaning with Obstacle Detection and Removal

November 2025 - Present (8 months)

B2B SaaS run by agentic AI. Scaled to 15+ clients across hospitality, real estate, SME finance & luxury retail, with ~\$10K MRR in 6 months.

### SolarMobil Manipal

Electronics Head

January 2024 - Present (2 years 6 months)

Manipal, India

- Implemented TouchGFX and FreeRTOS-based firmware for Cortex-M7 ECU and display units, improving thread latency by 210ms.
- Designed RF telemetry + weather/spatial fusion cloud dashboards for real-time race strategy, contributing to the best solar endurance performance in the country.
- Introduced version control and HIL pipelines to improve schedule adherence to 90%+

### Teachafy Labs

Machine Learning Engineer

July 2025 - December 2025 (6 months)

Bengaluru

Developed Google ADK-based agentic orchestration and WhatsApp business chatbots to inject natural language workflows into ERP platforms, saving 5000+ hrs/week across 17+ clients.

### Symbionic

Embedded Software Engineer

April 2025 - June 2025 (3 months)

Chennai, Tamil Nadu, India

Developed firmware around custom ESP32-based hardware for EMG/EEG signal acquisition & spatiotemporal fusion

### Manipal Institute of Technology

Undergraduate Research Assistant

April 2025 - June 2025 (3 months)

Manipal, India

Interfaced biomedical probes (ECG, temp, GSR) and innovative respiration measurement using IMUs with digital signal processing & IoT-derived sensor fusion for hospital management & biofeedback digital twin studies. Reduced hospital staff response times to <5 min, enhancing patient survivability by 35% +

## Foodswipe

### Co-Founder

January 2025 - June 2025 (6 months)

Bengaluru

Outsourcing food choices to content-based & collaborative recommender systems that actually understand what you'd like to eat. A/B testing rated positive customer experience at 85%+

## DCPR AI

### Software Engineer

July 2024 - December 2024 (6 months)

Bengaluru, Karnataka, India

Built document ingestion pipelines to automate PDF batch processing from GCS buckets using intelligent OCR inference at scale. Ingested 250TB+ poorly scanned docs through distributed Redis locking, concurrent Python workers, ELK monitoring, and Kubernetes HPA. Replaced manual processing, slashing TTM by 80%+ and saving \$600+/mo in data entry costs.

Developed RAG systems for legal and zoning documents with OpenAI Assistants API & Pinecone vector DB, achieving precise document & page citations with 92% accuracy, reducing legal compliance timelines for real estate developers in Mumbai by almost half.

## Krop AI

### Embedded Engineer

September 2024 - November 2024 (3 months)

Manipal, Udupi, Karnataka, India

Developed control & sensing systems + Jetson AI vision/perception pipeline for agritech systems.

## CoreEL Technologies

### Digital Design Engineer

April 2024 - June 2024 (3 months)

Bengaluru, Karnataka, India

Developed RTL design spec of high-speed CAN bus & I2C controllers, optimizing memory/macros for Zynq 7000 FPGA.

Collaborated with synthesis & verification teams to ensure QA standards were met. Integrated relevant IP blocks and gained expertise with the Vitis software suite.

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## Education

### Manipal Institute of Technology

Bachelor of Technology - BTech, Electrical, Electronics and Communications Engineering · (July 2023 - July 2027)

### St. Francis De Sales Public School

High School Diploma, Computer Science · (June 2017 - May 2023)

### Primrose Schools Chennai

Middle School, Science · (2010 - 2017)